Cairo University

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Software Engineering

Dental Clinic Management System

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Contents

1. INTRODUCTION

1.1 Background: Cephalometric Analysis in Orthodontics

Cephalometric analysis is a fundamental diagnostic tool in orthodontics and maxillofacial surgery, used to evaluate skeletal, dental, and soft-tissue relationships of the craniofacial complex. Traditionally, this process requires orthodontists to manually identify specific anatomical landmarks on a lateral cephalometric radiograph, connect them to construct geometric planes, and calculate angles and linear distances relative to healthy population norms.

1.2 Limitations of Manual Tracing

Manual tracing is time-consuming, highly subjective, and prone to significant inter-observer and intra-observer variability. Errors in landmark localization directly propagate to diagnostic parameters, potentially resulting in suboptimal treatment plans. Studies have shown that inter-observer variability in landmark identification can range from 0.5mm to over 2mm for key points such as Gonion and Articulare, leading to clinically significant discrepancies in derived measurements.

1.3 Proposed Solution: AI-Powered Cephalometric Analysis System

This project introduces the AI-Powered Cephalometric Analysis System, an enterprise-grade, end-to-end clinical workstation. The system integrates deep-learning keypoint regression, robust statistical uncertainty modeling, cervical vertebral maturation (CVM) growth forecasting, and a clean-architecture gateway. By automating landmark localization and clinical diagnostics, the platform enhances diagnostic consistency, saves valuable clinical time, and offers interactive, cloud-backed controls for clinical validation.

1.4 Dental Clinic Management System Overview

The Dental Clinic Management System is an integrated software solution designed to automate and optimize the operational workflow داخل عيادات الأسنان. The system addresses challenges related to manual data handling, inefficient appointment scheduling, and lack of structured medical record management.

1.5 Machine Learning Diagnostic Module

In addition to standard management functionalities, the system incorporates a Machine Learning–based diagnostic module capable of analyzing orthodontic X-ray images. This module enhances clinical decision-making by providing predictive insights derived from trained models.

1.6 Target Users and Intended Use

The system is intended for internal use by clinic staff and medical professionals, ensuring controlled access, data consistency, and improved service quality.

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2. SCOPE

1. System Workflow: Eight Integrated Stages

The system encompasses the entire clinical diagnostic and analysis workflow across eight integrated stages:

- **Intake & Upload:** Cloud-ready multi-tenant secure ingestion of lateral radiographs with patient profile management.
- **AI-Driven Localization:** HRNet-based predictions yielding sub-millimeter precision for 19 standard cephalometric landmarks.
- **Interactive Adjustment Cockpit:** Real-time dragging, zooming, and sub-pixel snapping to local intensity gradients on a custom HTML5 canvas.
- **Multi-Protocol Geometric Engine:** Automatic evaluation of 10 standard clinical protocols (Steiner, Tweed, Downs, McNamara, Jarabak, and others) dynamically adapted to patient demographic groups.
- **Skeletal Staging (CVM):** Cervical vertebral segmentation forecasting residual growth potential and optimal clinical treatment windows.
- **Clinical Narrative Generators:** LLM-driven generation of warm letters for patients and structured technical descriptions for clinicians.
- **QuestPDF Export & Case Archiving:** Automated export of diagnostic envelopes containing PDF reports, raw values, and WeDoCeph-compatible ZIP configurations.
- **System Integrations:** OpenTelemetry distributed telemetry, Docker containerization, and Prometheus/Grafana monitoring.

2. Dental Clinic Management Capabilities

This system focuses on supporting the internal operations of a dental clinic through the following capabilities:

- Patient data management (registration, update, retrieval)
- Appointment scheduling, modification, and tracking
- Medical record creation and history tracking
- Integration of a Machine Learning model for orthodontic X-ray analysis

3. System Access Model

The system operates as a closed internal system, meaning that patients do not directly interact with it. All interactions are handled بواسطة staff and doctors.

4. Out of Scope

The following features are out of scope:

- Online booking by patients
- Payment processing
- External system integration

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3. PURPOSE

1. Addressing Critical Gaps in Current Orthodontic Practices

The platform addresses four critical gaps in current orthodontic practices:

- **Elimination of Subjective Variance:** Stabilizes landmark positioning using a model trained on large clinical datasets, delivering high diagnostic consistency across practitioners and institutions.
- **Immediate Geometric Outputs:** Calculates skeletal, vertical, and dental vectors instantly, reducing patient appointment turnaround from days to minutes.
- **Accurate Growth Forecasts:** Integrates CVM skeletal maturation indexing to ensure orthopedic appliances are applied during the optimal pubertal growth window.
- **Patient-Centered Explanations:** Uses generative AI to translate complex orthodontic measurements into empathetic, plain-language patient letters, improving treatment compliance and patient satisfaction.

2. Primary Purpose of the System

The primary purpose of this system is to:

- Improve operational efficiency داخل العيادة
- Reduce manual errors associated with paper-based systems
- Provide a structured and scalable data management solution

- Support medical decision-making باستخدام Machine Learning

3. Strategic Objective

The integration of AI-driven analysis aims to bridge the gap between traditional clinic systems and intelligent healthcare solutions.

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4. SYSTEM ARCHITECTURE

4.1 Layered Architecture Overview

The system follows a layered architecture consisting of:

- Presentation Layer (Frontend UI)
- Application Layer (Backend API & Business Logic)
- Data Layer (Database)
- Machine Learning Module (X-ray analysis)

4.2 Microservice-Oriented Design

The AI Cephalometric Analysis System implements a microservice-oriented design distributed across four primary functional tiers:

- **Presentation Tier:** Vite 7 + React 19 Client SPA with Zustand Canvas Context and i18n internationalization support.
- **API Gateway Tier:** ASP.NET Core Web API Gateway with CQRS (Mediator) pattern, Serilog structured logging, and JWT authentication.
- **AI Service Tier:** FastAPI Web Engine with PyTorch HRNet Core, CVM & Monte-Carlo Simulation modules, and SQLite3 Local Case Repository.
- **Persistence Tier:** PostgreSQL for cloud profiles, Redis distributed cache, S3/Local storage modules, and OpenTelemetry Collector for observability.

5. SYSTEM REQUIREMENTS SPECIFICATION

5.1 Functional Requirements (Detailed)

ID	Requirement	Expected Behavior
FR-1	User Management	Secure sign-in, registration, email verification, role-based access
FR-2	Patient Profile CRUD	Validated operations for MRN, age, sex, study history
FR-3	Radiograph Ingestion	High-resolution lateral cephalograms with calibration
FR-4	AI Inference	19-landmark detection with HRNet-W32
FR-5	Coordinate Refinement	Sub-pixel snapping on HTML5 canvas
FR-6	Diagnostic Engine	Skeletal, vertical, dental measurements + CVM staging
FR-7	PDF Reporting	QuestPDF with confidence intervals and outlier flags
FR-8	GenAI Narratives	Clinician summaries and patient-friendly letters
FR-9	Export & Archiving	ZIP bundles and SQLite case repository

Role-Based Functional Requirements

Staff:

- Register patients
- Manage appointments

Doctor:

- View appointments
- Add medical records
- Analyze X-rays

5.2 System Requirements Specification

The system must:

- Store and manage patient data
- Handle appointment scheduling
- Maintain medical records
- Integrate ML model for image analysis
- Support role-based access control

5.3Requirement Specifications

- The system shall allow staff to register patients
- The system shall allow doctors to add medical records
- The system shall process X-ray images and return results

5.4User Stories

Actors

Actor	Description	Responsibilities
Staff	Clinic receptionist	Patient registration, appointment management, data entry
Doctor	Orthodontic specialist	X-ray upload, medical documentation, diagnosis
System	Automated processes	Data validation, workflow integration,AI coordination

Priority Distribution

Priority	User Stories	Count	Percentage
High (Must) 🔴	US-01, US-02, US-03, US-05, US-06, US-07	6	60%
Medium (Should) 🟡	US-04, US-08, US-09	3	30%
Low (Could) 🟢	US-10	1	10%

User Stories Details

US-01: Patient Management

Priority : High 🔴

Actor : Staff

Story : As a staff member, I want to register and manage patient information, including personal and contact details, so that all patient records are accurate, organized, and easily accessible when needed.

Acceptance Criteria:

- Staff can add new patient (Name, Age, Gender, Phone)
- System validates phone number format
- System auto-generates unique PatientID
- Staff can edit existing patient information
- Staff can search by name or ID
- Staff can view all patients list
- System prevents duplicate entries

Dependencies : None (Foundation)

US-02: Appointment Scheduling Workflow

Priority : High 🔴

Actor : Staff

Story : As a staff member, I want to schedule, update, and cancel appointments efficiently based on doctor availability, so that patient flow داخل العيادة يكون منظم بدون تعارضات أو تأخير.

Acceptance Criteria:

- Staff can book appointment for existing patient
- Staff can select doctor and time slot
- System shows only available slots
- System prevents double-booking
- Staff can update appointment (date, time, status)
- Staff can cancel appointment
- Staff can view daily/weekly schedule

Dependencies : US-01

US-03: Patient Record Accessibility

Priority : High 🔴

Actor : Doctor

Story : As a doctor, I want to access complete patient records, including history and previous diagnoses, so that I can make informed medical decisions بسرعة ودقة.

Acceptance Criteria:

- Doctor can view patient personal information
- Doctor can view appointment history
- Doctor can view medical records history
- Doctor can view X-ray images
- Doctor can view previous AI results
- View-only mode for existing data

Dependencies : US-01, US-04

US-04: Medical Documentation

Priority : Medium 🟡

Actor : Doctor

Story : As a doctor, I want to create and update medical records after each consultation, so that patient progress is tracked and documented بشكل احترافي.

Acceptance Criteria:

- Doctor can add new medical record
- Record includes (Date, Diagnosis, Notes, Treatment)
- Doctor can edit existing records
- System timestamps all changes
- Doctor name recorded with each entry
- Records linked to specific appointment

Dependencies : US-01

US-05: X-ray Image Management

Priority : High 🔴

Actor : Doctor

Story : As a doctor, I want to upload and manage patient X-ray images داخل النظام، so that all diagnostic data يكون متخزن في مكان واحد وسهل الرجوع له.

Acceptance Criteria:

- Doctor can upload X-ray for specific patient
- System accepts JPEG, PNG, DICOM formats
- File size limit: 10MB
- Image stored with PatientID and date
- Doctor can view all images per patient
- Doctor can delete image with confirmation

Dependencies : US-01

US-06: AI-Based Image Analysis

Priority : High 🔴

Actor : Doctor / System

Story : As a doctor, I want the system to automatically analyze uploaded X-ray images using a machine learning model, so that I receive accurate cephalometric measurements without manual effort.

Acceptance Criteria:

- System auto-sends image to AI after upload
- AI processes using trained ML model
- AI returns measurements and predictions
- AI returns confidence score (0-100%)
- Processing time ≤ 5 seconds
- Error handling for AI failure

Dependencies : US-05

US-07: AI-Assisted Decision Making

Priority : High 🔴

Actor : Doctor

Story : As a doctor, I want to view AI-generated analysis results alongside patient data, so that I can combine clinical expertise with intelligent system support to improve diagnosis accuracy.

Acceptance Criteria:

- Display AI results with patient data
- Show confidence score visually (progress bar)
- Doctor can accept or flag AI results
- Compare AI results with manual assessment

Dependencies : US-06

US-08: Data Consistency and Integrity

Priority : Medium 🟡

Actor : System

Story : As a system, I want to ensure that all patient, appointment, and analysis data are stored consistently and securely, so that data integrity is maintained across all operations.

Acceptance Criteria:

- Enforce foreign key relationships
- No orphan records
- Daily backups
- Audit log for all changes

Dependencies : All entity tables

US-09: Automated Workflow Integration

Priority : Medium 🟡

Actor : System

Story : As a system, I want to integrate appointment management with medical analysis processes, so that the workflow from patient registration to diagnosis is seamless and efficient.

Acceptance Criteria:

- After appointment → upload X-ray
- After X-ray → AI analysis auto-triggers
- After AI → add medical record
- System tracks workflow state

Dependencies : US-02, US-05, US-06, US-04

US-10: Error Handling and Validation

Priority : Low 🟢

Actor : System

Story : As a system, I want to validate inputs (e.g., patient data, uploaded images) before processing, so that errors are minimized and system reliability is improved.

Acceptance Criteria:

- Validate phone number format
- Validate age range (0-120)
- Validate image format and size
- User-friendly error messages in Arabic

Dependencies : None

5.5 Work Backlog

Tasks are distributed across 4 sprints. Total duration: **12 weeks**.

Sprint 1: Foundation (Weeks 1-3)

ID	Task Name	Description	Assigned To	Priority	Status
T01	Database Design	Design Patient, Appointment, XrayImage, AnalysisResult tables	Eng / Abdelrahman	High	DONE
T02	Patient Model	CRUD operation for patient (add, edit, search ,delete)	Eng / Ahmed	High	DONE
T03	Patient UI	Patient management interface	Eng / Rania	High	DONE

Sprint 2: Core Operations (Weeks 4-6)

ID	Task Name	Description	Assigned To	Priority	Status
T04	Appointment Module	Booking, update, cancel with availability check	Eng / Ahmed	High	DONE
T05	Appointment UI	Calendar view, booking forms	Eng / Ahmed	High	DONE
T06	Doctor Module	Doctor login, profile, schedule views	Eng / Rania	High	DONE
T07	Patient Record View	View patient history with appointments and records	Eng / Ahmed	High	DONE

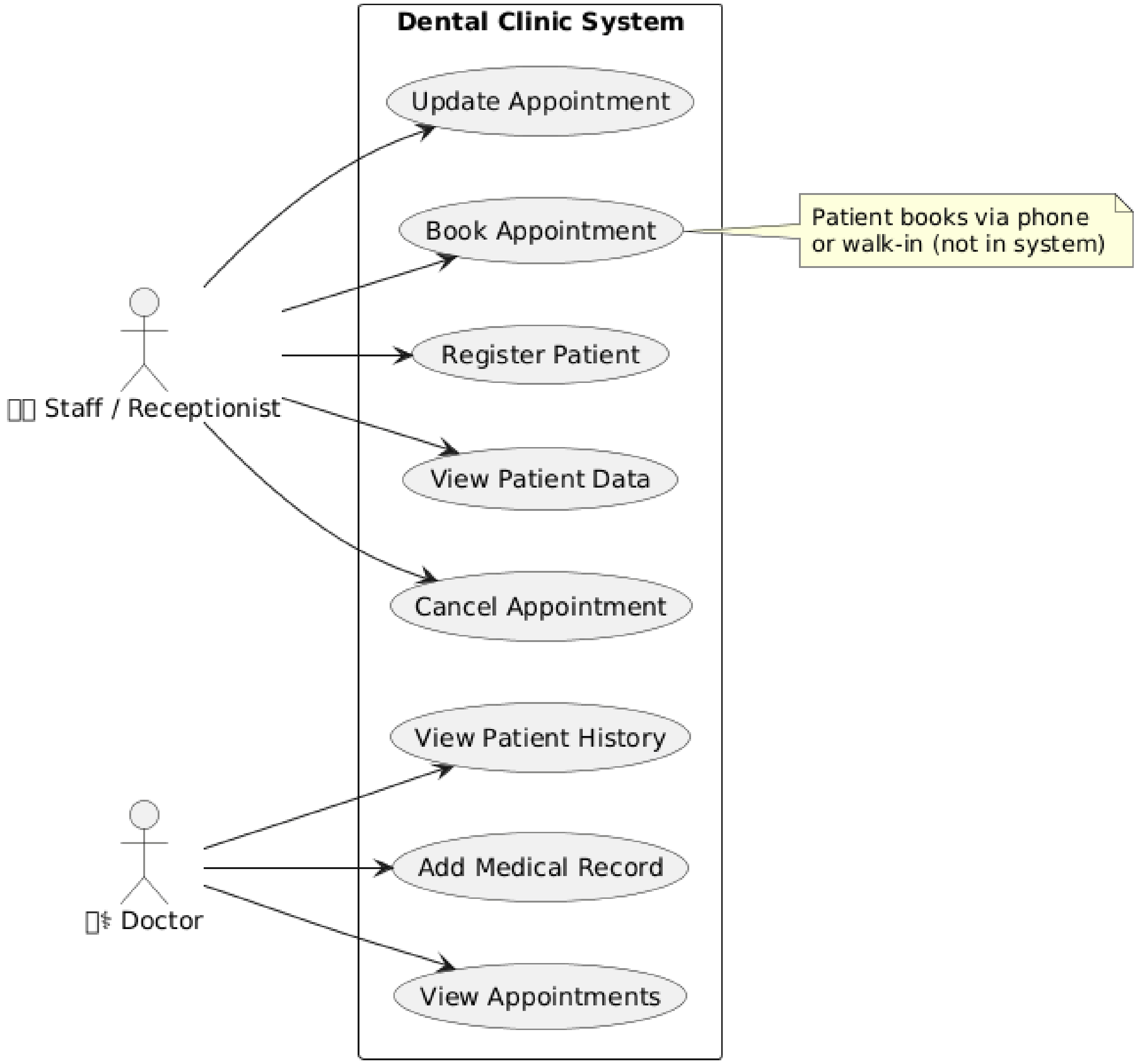
Sprint 3: AI Integration (Weeks 7-9)

ID	Task Name	Description	Assigned To	Priority	Status
T12	AI Results Display	Display AI results with confidence score	Eng / Abdelrahman	High	DONE
T13	Workflow Integration	Connect appointment → X-ray → AI → medical record	Eng / Abdelrahman	Medium	DONE
T14	Error Handling	Input validation and Arabic error messages	Eng / Ahmed	low	DONE
T15	System Testing	Unit testing, integration testing, UAT	Eng / Rania	Medium	DONE
T16	Deployment	Deploy to server, backup and security	Eng / Ahmed	Medium	DONE

5.6Project Duration

- Analysis Phase
- Design Phase
- Implementation Phase
- Testing Phase

5.7Use Case Diagram



5.8 Actor Description

- **Staff / Receptionist**

The staff (receptionist) is responsible for managing the administrative operations داخل النظام.
He/She interacts directly with patients (outside the system) and uses the system to perform the following tasks:

- Register new patients
- Book appointments (via phone or walk-in)
- Update appointment details
- Cancel appointments
- View patient data

The staff acts as the main interface between patients and the system.

- **Doctor**

The doctor is responsible for the medical الجانب داخل النظام.
He/She uses the system to access patient information and manage medical records.

The doctor can:

- View patient history
- Add medical records
- View appointments

The doctor uses the system mainly for diagnosis and patient follow-up.

- **AI System (Machine Learning Model)**

The AI System represents the machine learning model responsible for analyzing cephalometric X-ray images.

It interacts with the system automatically (not a human actor) and performs the following:

- Receives uploaded X-ray images from the doctor
- Processes the image using trained ML algorithms
- Generates analysis results (measurements, predictions)
- Sends results back to the system for storage and display

This actor is considered a system component that enhances decision-making.

5.9Use Case Description

- **Register Patient**

Actor	**Staff**
Description	The staff enters patient information into the system
Precondition	Patient is not already registered
Postcondition	Patient data is stored in the database

- **Book Appointment**

Actor	Staff
Description	The staff books an appointment for a patient (via phone or walk-in)
Precondition	Patient exists in the system
Postcondition	Appointment is saved

- **Update Appointment**

Actor	Staff
Description	The staff modifies appointment details (date/time)
Precondition	Appointment exists
Postcondition	Appointment is updated

- **Cancel Appointment**

Actor	Staff
Description	The staff cancels an existing appointment
Precondition	Appointment exists
Postcondition	Appointment is removed or marked as cancelled

- **View Patient Data**

Actor	Staff
Description	The staff views patient information
Precondition	Patient exists
Postcondition	Patient data is displayed

- **View Patient History**

Actor	Doctor
Description	The doctor views previous medical records of the patient
Precondition	Patient exists
Postcondition	History is displayed

- **Add Medical Record**

Actor	Doctor
Description	The doctor adds diagnosis or notes to patient record
Precondition	Patient exists
Postcondition	Record is stored

- **View Appointments**

Actor	Doctor
Description	The doctor views scheduled appointments
Precondition	Appointments exist
Postcondition	Appointments list is displayed

- **Upload X-ray Image**

Actor	Doctor
Description	The doctor uploads a patient's X-ray image into the system
Precondition	Patient exists
Postcondition	Image is stored and ready for analysis

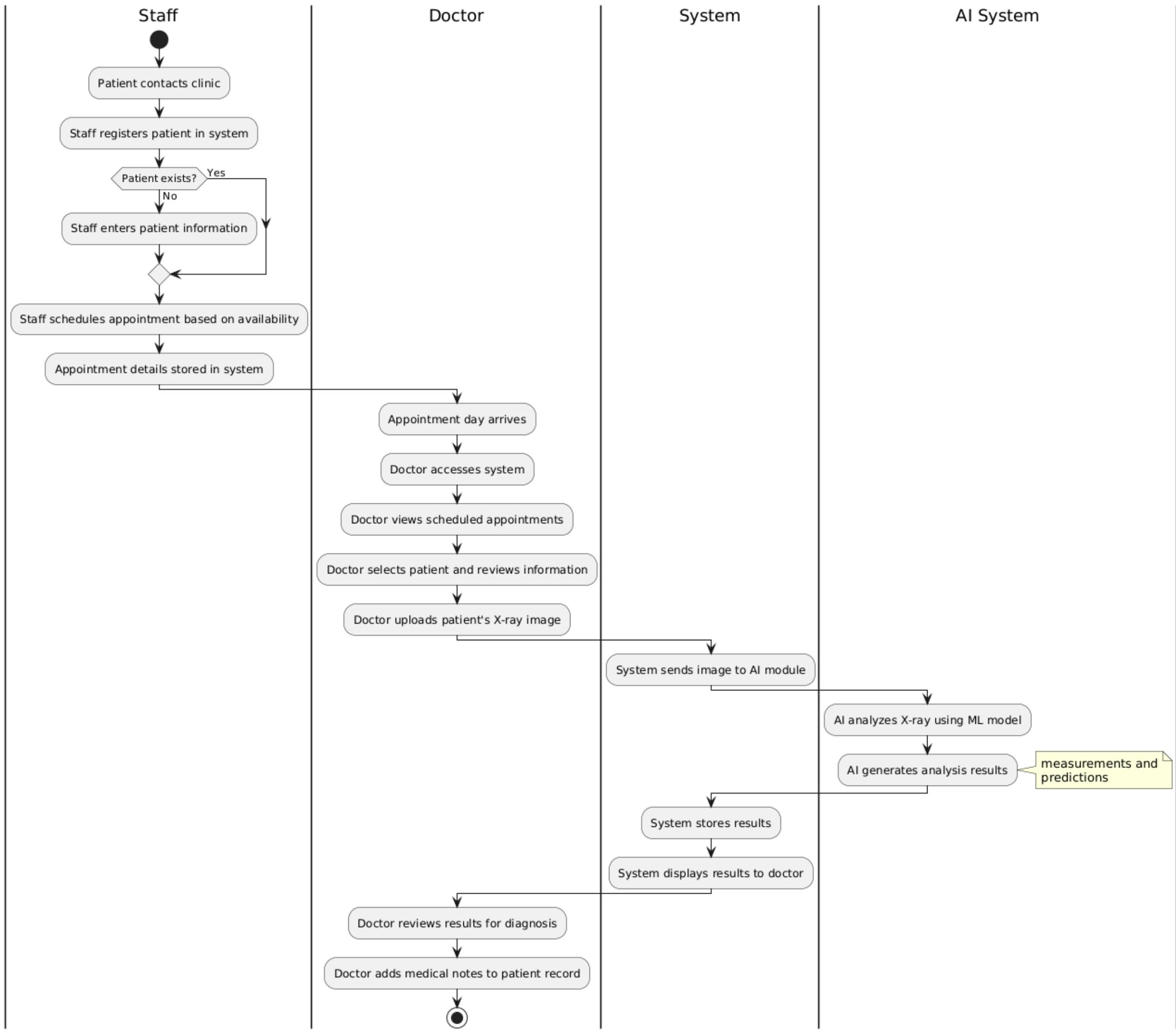
- **Analyze X-ray Image**

Actor	AI System
Description	The system automatically processes the uploaded image using a machine learning model
Precondition	X-ray image is uploaded
Postcondition	Analysis result is generated

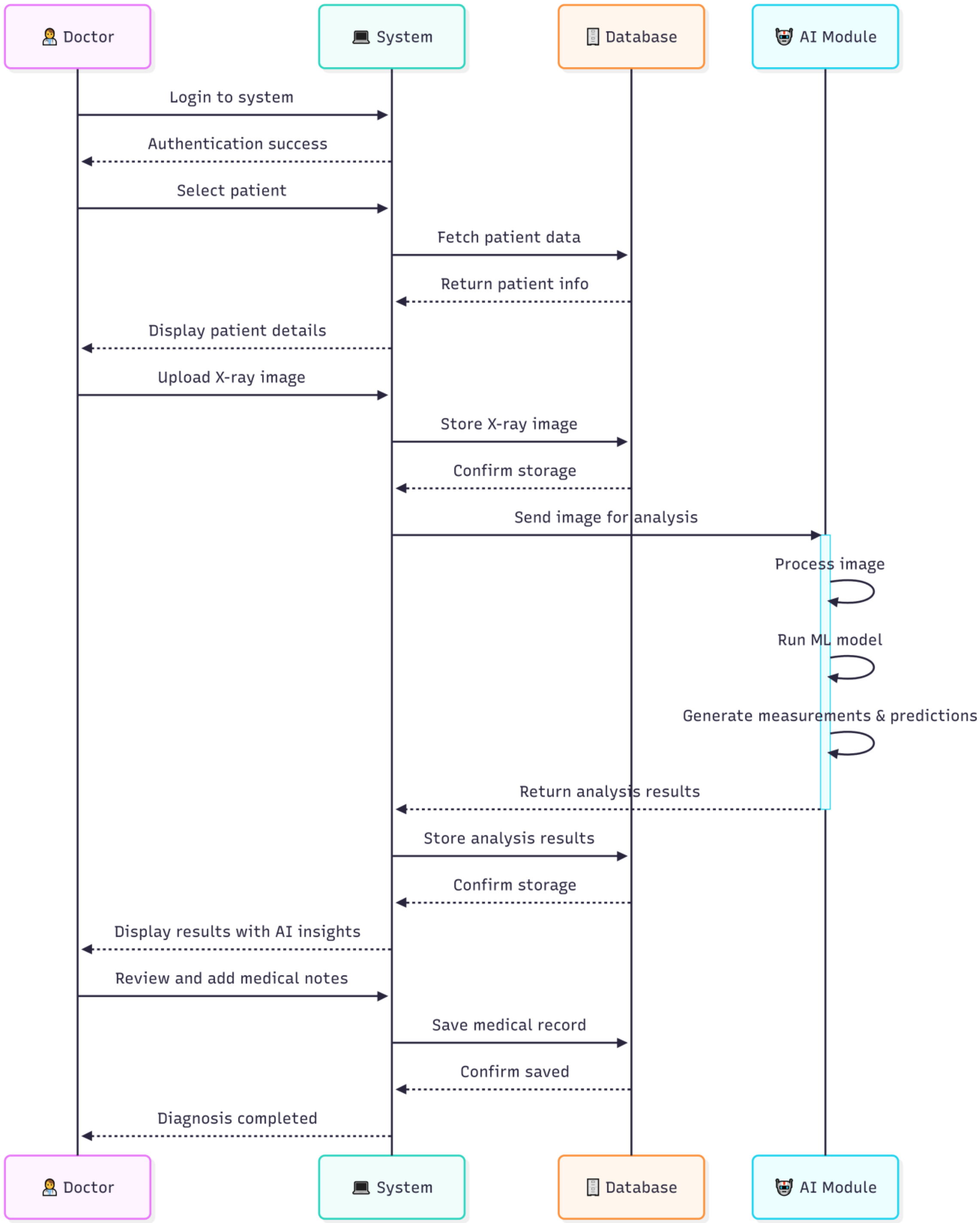
- **View Analysis Result**

Actor	Doctor
Description	The doctor views the AI-generated analysis results
Precondition	Analysis is completed
Postcondition	Results are displayed to assist diagnosis

5.10 Activity Diagrams



5.11 Sequence Diagrams



6. NON-FUNCTIONAL REQUIREMENTS

Overview

The system must address the following non-functional quality attributes:

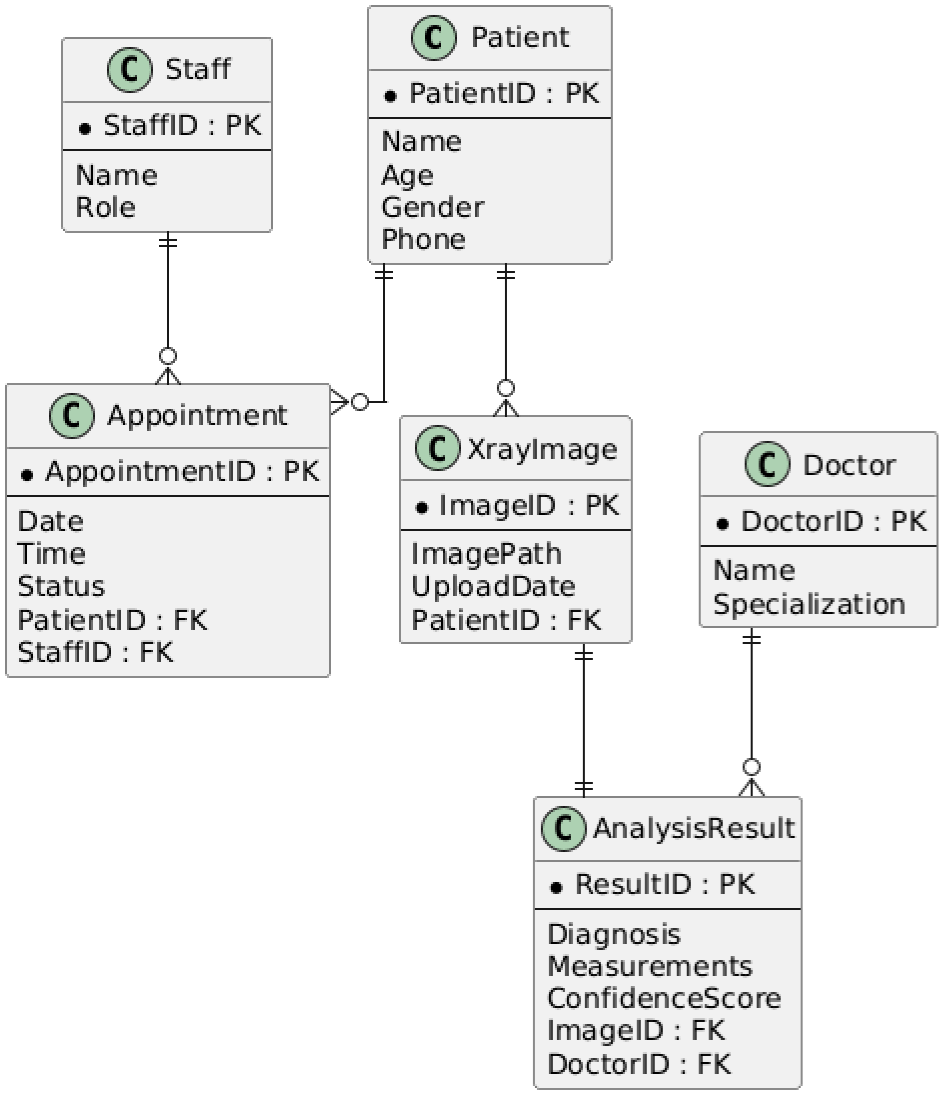
- Security
- Performance
- Usability
- Reliability

Measurable Targets

ID	Requirement	Target / Measure
NFR-1	Spatial Precision	Mean radial error < 1.0 mm
NFR-2	Pipeline Latency	End-to-end < 5.0 seconds
NFR-3	Scalability	100+ concurrent users
NFR-4	Availability	99.9% service uptime
NFR-5	Security	AES-256 at rest, TLS 1.3 in transit
NFR-6	Visual Performance	60 FPS canvas interaction

7. SYSTEM DESIGN

Entity Relationship Diagram (ERD)



Mapping

- Each entity converted to table
- Primary keys assigned
- Foreign keys used for relationships
- AI results stored in separate table

8. TESTING (USER ACCEPTANCE TEST)

in progress